

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1
install.packages("tidyverse")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'
## (as 'lib' is unspecified)
```

```
install.packages("agricolae")
```

```
## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'
## (as 'lib' is unspecified)
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    4.0.0      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.0.2

## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
library(ggplot2)

getwd()
```

```
## [1] "/home/guest/GITclone/EDE_Fall2025"
```

```
Lake_ChemPhys_Raw <-
  read.csv("/home/guest/GITclone/EDE_Fall2025/Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv",
           stringsAsFactors = TRUE)

Lake_ChemPhys_Raw$sampldate <-
  as.Date(Lake_ChemPhys_Raw$sampldate,
          format = "%m/%d/%y")

head(Lake_ChemPhys_Raw$sampldate)
```

```
## [1] "1984-05-27" "1984-05-27" "1984-05-27" "1984-05-27" "1984-05-27"
## [6] "1984-05-27"
```

```
#2
my_theme <-
  theme(
    plot.background = element_rect(fill = "white"),
    legend.background = element_rect(fill = "white"),
    legend.title = element_text(color = "turquoise4",
                                 face = "bold"),
    legend.text = element_text(color = "turquoise4"),
    panel.background = element_rect(fill = "white"),
    panel.grid.major = element_line(color = "turquoise4"),
    panel.grid.minor = element_line(color = "turquoise4"),
    plot.title = element_text(face = "bold",
                              color = "turquoise4"),
    axis.title.x = element_text(
      face = "italic",
      size = 14,
      color = "turquoise4"),
    axis.title.y = element_text(
      face = "italic",
      size = 14,
      color = "turquoise4"),
  )
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: Mean lake temperatures recorded during July does not change with depth across all lakes. Ha: Mean lake temperatures recorded during July change with depth across all lakes.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
Lake_ChemPhys <-
  Lake_ChemPhys_Raw%>%
  subset(month(sampledate)==7)%>%
  select(lakename,
         year4,
         daynum,
         depth,
         temperature_C)%>%
  drop_na()

#5
library(viridis)
```

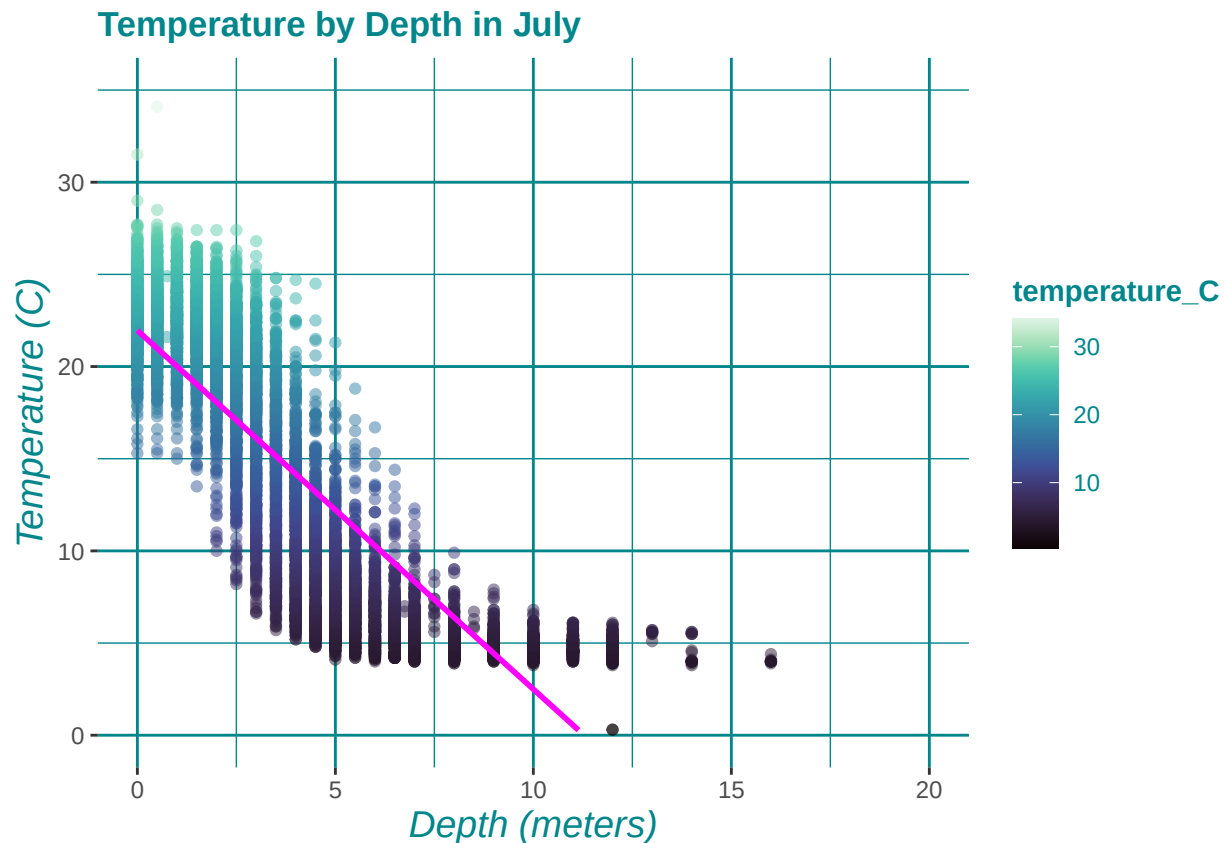
```
## Loading required package: viridisLite
```

```
Lake_ChemPhys_Plot <-
  ggplot(Lake_ChemPhys,
         aes(x = depth,
             y = temperature_C,
             color = temperature_C)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm",
             se = TRUE,
             color = "magenta") +
  xlim(0,20)+
  ylim(0,35)+
  labs(
    title = "Temperature by Depth in July",
    x = "Depth (meters)",
    y = "Temperature (C)") +
  scale_color_viridis(option = "mako", direction = 1) +
  my_theme

print(Lake_ChemPhys_Plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The figure suggests that in July, as depth increases, temperature decreases within the lake. The lowest temperatures plateau at around 4 degrees C. So, at a certain point, depth doesn't have any increased impact on the temperature of the water.

7. Perform a linear regression to test the relationship and display the results.

```
#7
tempdepth.regression <- lm(
  data = Lake_ChemPhys,
  temperature_C ~ depth)

summary(tempdepth.regression)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth, data = Lake_ChemPhys)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 21.95597    0.06792   323.3  <2e-16 ***
## depth       -1.94621    0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: With an R-squared value of 0.7387, about 73.87% of the variability in temperature is explained by changes in depth. This finding is based on 9726 degrees of freedom which means that there was a large, sufficient sample size to observe this relationship. These results are also statistically significant with a p-value of $<2.2e-16$, which is far within the threshold of statistical significance: 0.05. With every 1m increase in depth, temperature decreases by around 1.94621 degrees C.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

```
#9
library(dplyr)
install.packages("corrplot")

## Installing package into '/home/guest/R/x86_64-pc-linux-gnu-library/4.4'
## (as 'lib' is unspecified)

library(corrplot)

## corrplot 0.95 loaded
```

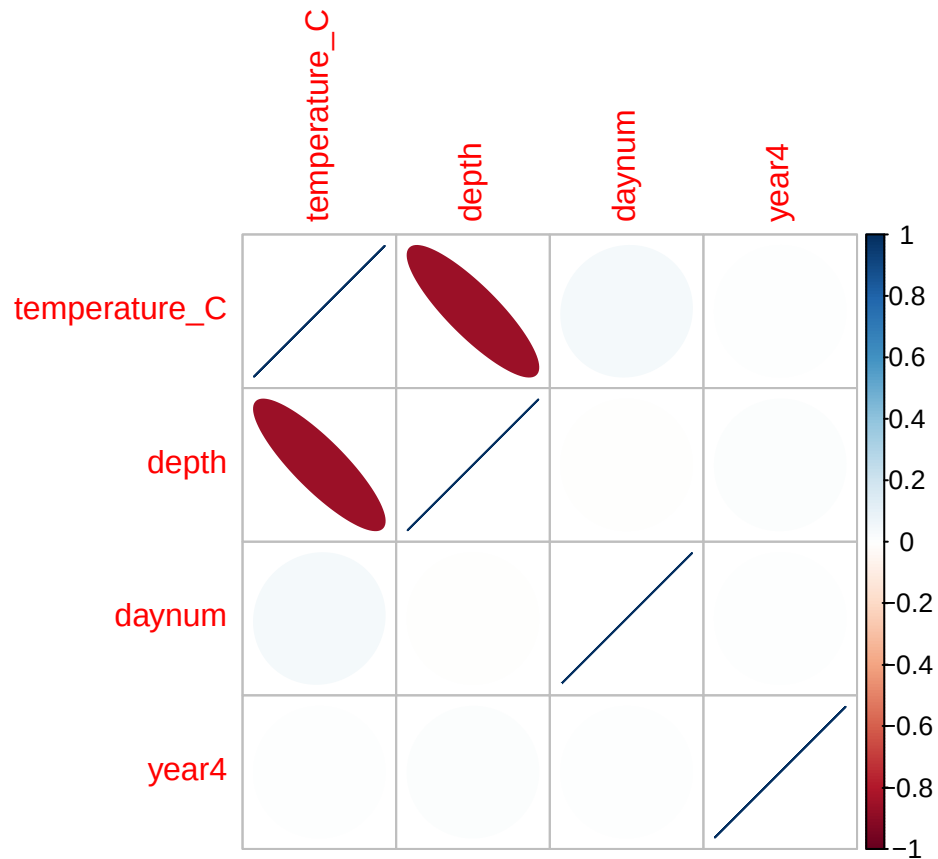
```

Lake_ChemPhys_sub <-
Lake_ChemPhys %>%
  select(temperature_C:year4) %>%
  na.omit()

Lake_ChemPhys_Corr <- cor(Lake_ChemPhys_sub)

corrplot(Lake_ChemPhys_Corr,
  method = "ellipse")

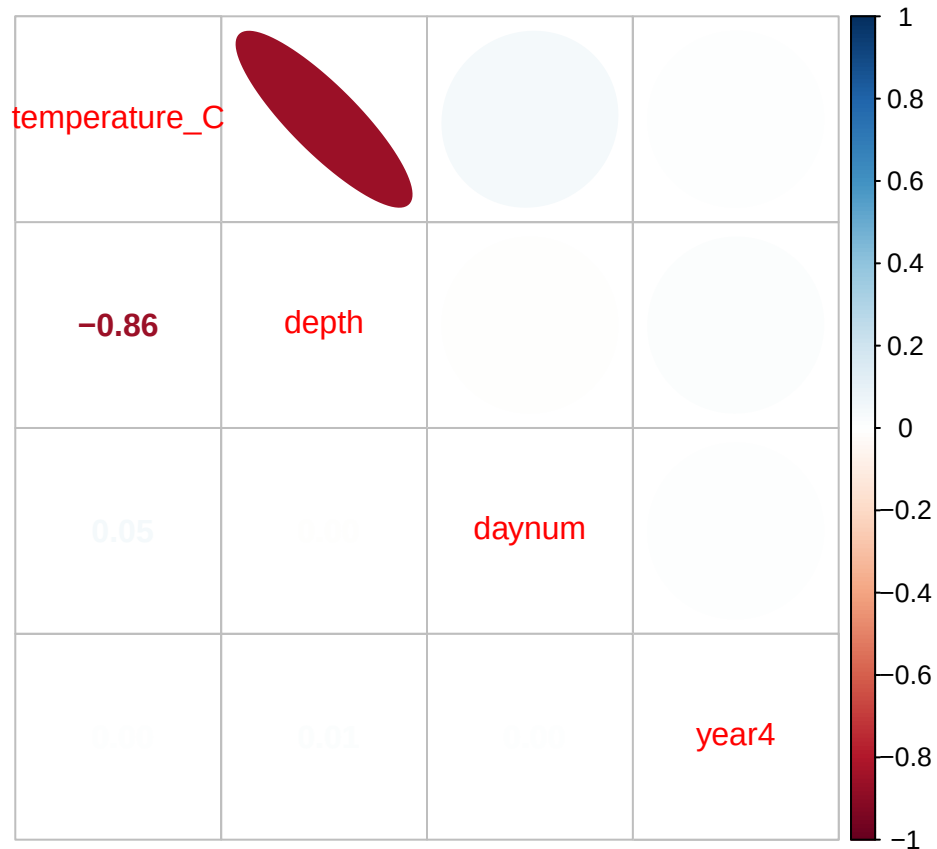
```



```

corrplot.mixed(Lake_ChemPhys_Corr,
  upper = "ellipse")

```



#10

```
Lake_ChemPhys_BestRegression<- lm(data = Lake_ChemPhys_sub,
temperature_C ~ depth + daynum)
```

```
summary(Lake_ChemPhys_BestRegression)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth + daynum, data = Lake_ChemPhys_sub)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6174 -2.9809  0.0845  2.9681 13.4406
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 14.088588  0.855505  16.468  <2e-16 ***
## depth       -1.946111  0.011685 -166.541 <2e-16 ***
## daynum        0.039836  0.004318   9.225  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.818 on 9725 degrees of freedom
## Multiple R-squared:  0.741, Adjusted R-squared:  0.741
## F-statistic: 1.391e+04 on 2 and 9725 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression are depth and daynum. This model explains 74.1% of the observed variance with an R-squared value of 0.741. This is a slight improvement over the model using only depth, as that previous model could explain ~0.23% less of the observed variance.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
Lake_ChemPhys_ANOVA <- aov(data = Lake_ChemPhys,
                           temperature_C ~ lakename)
summary(Lake_ChemPhys_ANOVA)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## lakename      8  21642   2705.2     50 <2e-16 ***
## Residuals  9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Lake_ChemPhys_LM <- lm(data = Lake_ChemPhys,
                      temperature_C ~ lakename)
summary(Lake_ChemPhys_LM)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = Lake_ChemPhys)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    17.6664    0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145    0.7699  -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987    0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake -6.8931    0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake       -3.8522    0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake      -4.3501    0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake    -6.5972    0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake       -3.2078    0.9429  -3.402 0.000672 ***
```

```
## lakenameWest Long Lake    -6.0878    0.6895   -8.829   < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

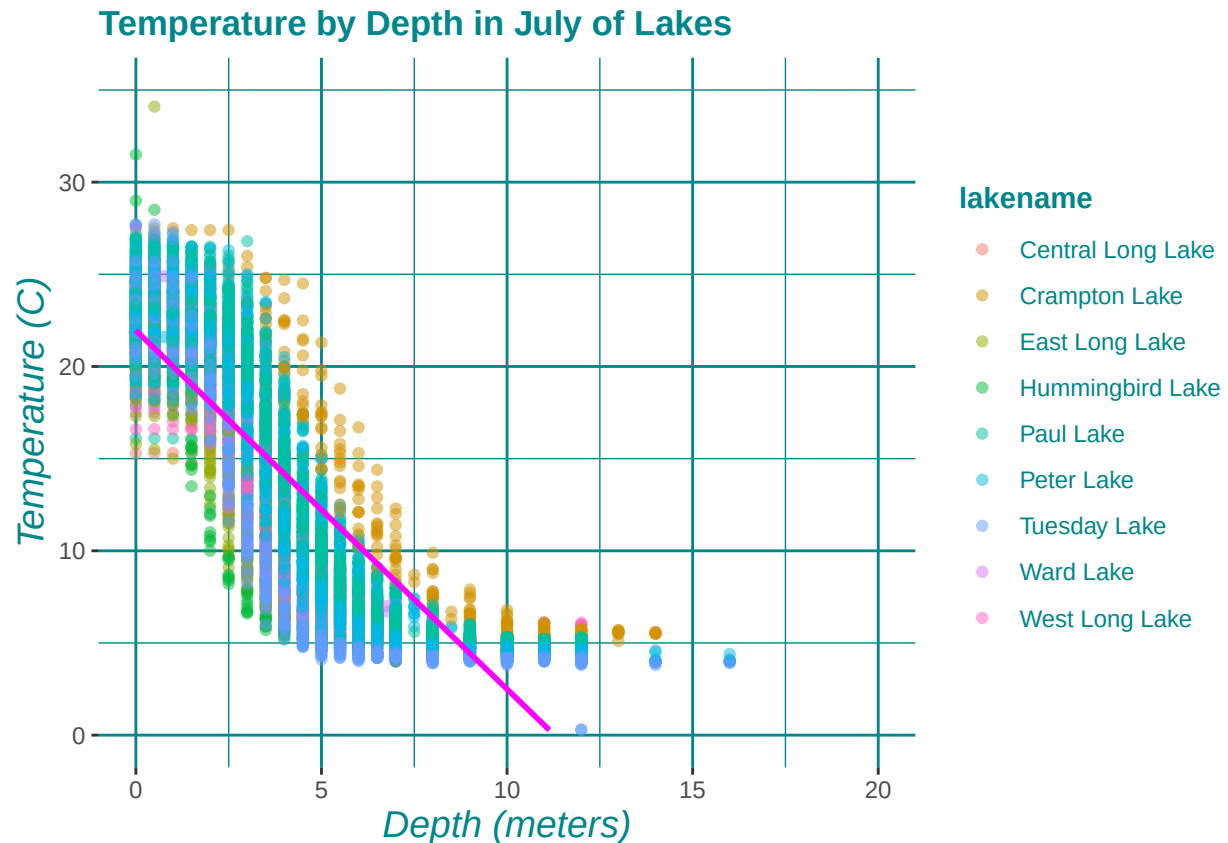
Answer: Yes, there is a significant difference in mean temperature among the lakes. From both the linear model and ANOVA test, the adjusted p-value for the difference in mean temperature among the lakes is statistically significant at $<2e-16$ (much less than 0.05 or 0.001).

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
Lake_ChemPhys_Plot2 <-
  ggplot(Lake_ChemPhys,
    aes(x = depth,
        y = temperature_C,
        color = lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm",
    se = TRUE,
    color = "magenta") +
  xlim(0,20)+
  ylim(0,35)+
  labs(
    title = "Temperature by Depth in July of Lakes",
    x = "Depth (meters)",
    y = "Temperature (C)") +
  my_theme
print(Lake_ChemPhys_Plot2)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

#15

```
TukeyHSD(Lake_ChemPhys_ANOVA)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = Lake_ChemPhys)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999
Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000

## Ward Lake-Crampton Lake	-0.8932661	-3.3684639	1.5819317	0.9714459
## West Long Lake-Crampton Lake	-3.7732318	-5.2378351	-2.3086285	0.0000000
## Hummingbird Lake-East Long Lake	0.5056106	-1.7364925	2.7477137	0.9988050
## Paul Lake-East Long Lake	3.5465903	2.6900206	4.4031601	0.0000000
## Peter Lake-East Long Lake	3.0485952	2.2005025	3.8966879	0.0000000
## Tuesday Lake-East Long Lake	0.8015604	-0.1363286	1.7394495	0.1657485
## Ward Lake-East Long Lake	4.1909554	1.9488523	6.4330585	0.0000002
## West Long Lake-East Long Lake	1.3109897	0.2885003	2.3334791	0.0022805
## Paul Lake-Hummingbird Lake	3.0409798	0.8765299	5.2054296	0.0004495
## Peter Lake-Hummingbird Lake	2.5429846	0.3818755	4.7040937	0.0080666
## Tuesday Lake-Hummingbird Lake	0.2959499	-1.9019508	2.4938505	0.9999752
## Ward Lake-Hummingbird Lake	3.6853448	0.6889874	6.6817022	0.0043297
## West Long Lake-Hummingbird Lake	0.8053791	-1.4299320	3.0406903	0.9717297
## Peter Lake-Paul Lake	-0.4979952	-1.1120620	0.1160717	0.2241586
## Tuesday Lake-Paul Lake	-2.7450299	-3.4781416	-2.0119182	0.0000000
## Ward Lake-Paul Lake	0.6443651	-1.5200848	2.8088149	0.9916978
## West Long Lake-Paul Lake	-2.2356007	-3.0742314	-1.3969699	0.0000000
## Tuesday Lake-Peter Lake	-2.2470347	-2.9702236	-1.5238458	0.0000000
## Ward Lake-Peter Lake	1.1423602	-1.0187489	3.3034693	0.7827037
## West Long Lake-Peter Lake	-1.7376055	-2.5675759	-0.9076350	0.0000000
## Ward Lake-Tuesday Lake	3.3893950	1.1914943	5.5872956	0.0000609
## West Long Lake-Tuesday Lake	0.5094292	-0.4121051	1.4309636	0.7374387
## West Long Lake-Ward Lake	-2.8799657	-5.1152769	-0.6446546	0.0021080

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Ward Lake and Peter Lake have the same mean temperature, statistically speaking, with a p-value of 0.7827, which is not statistically significant. Paul Lake and Peter Lake also have the same mean temperature with a statistically insignificant p-value of 0.2242. There is no lake that has a mean temperature that is statistically distinct from all other lakes. Each lake has at least one statistically insignificant pairing with another lake (p-value greater than 0.05).

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: If we were just looking at Peter and Paul Lake, another test we might explore to see whether they have distinct mean temperatures is a two-sample t-test.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
ChemPhys_Crampton_Ward <-
  Lake_ChemPhys%>%
  filter(lakename %in% c("Crampton Lake", "Ward Lake"))

ChemPhys_Crampton_Ward
```

```
##      lakename year4 daynum depth temperature_C
## 1   Crampton Lake 1999   196   0.00         22.8
```

## 2	Crampton Lake	1999	196	0.50	22.6
## 3	Crampton Lake	1999	196	1.00	22.4
## 4	Crampton Lake	1999	196	1.50	22.2
## 5	Crampton Lake	1999	196	2.00	22.0
## 6	Crampton Lake	1999	196	2.50	21.9
## 7	Crampton Lake	1999	196	3.00	21.7
## 8	Crampton Lake	1999	196	3.50	21.3
## 9	Crampton Lake	1999	196	4.00	20.0
## 10	Crampton Lake	1999	196	4.50	17.1
## 11	Crampton Lake	1999	196	5.00	14.4
## 12	Crampton Lake	1999	196	5.50	12.1
## 13	Crampton Lake	1999	196	6.00	10.2
## 14	Crampton Lake	1999	196	6.50	9.1
## 15	Crampton Lake	1999	196	7.00	8.4
## 16	Crampton Lake	1999	196	8.00	7.1
## 17	Crampton Lake	1999	196	9.00	6.3
## 18	Crampton Lake	1999	196	10.00	5.8
## 19	Crampton Lake	1999	196	11.00	5.5
## 20	Crampton Lake	1999	196	12.00	5.4
## 21	Crampton Lake	2003	183	0.00	21.8
## 22	Crampton Lake	2003	183	0.50	21.9
## 23	Crampton Lake	2003	183	1.00	21.9
## 24	Crampton Lake	2003	183	1.50	21.8
## 25	Crampton Lake	2003	183	2.00	21.8
## 26	Crampton Lake	2003	183	2.50	21.3
## 27	Crampton Lake	2003	183	3.00	20.8
## 28	Crampton Lake	2003	183	3.50	19.9
## 29	Crampton Lake	2003	183	4.00	17.2
## 30	Crampton Lake	2003	183	4.50	15.2
## 31	Crampton Lake	2003	183	5.00	12.7
## 32	Crampton Lake	2003	183	5.50	11.2
## 33	Crampton Lake	2003	183	6.00	10.2
## 34	Crampton Lake	2003	183	6.50	8.8
## 35	Crampton Lake	2003	183	7.00	8.2
## 36	Crampton Lake	2003	183	8.00	6.8
## 37	Crampton Lake	2003	183	9.00	6.0
## 38	Crampton Lake	2003	183	10.00	5.4
## 39	Crampton Lake	2003	183	11.00	5.1
## 40	Crampton Lake	2003	183	12.00	5.0
## 41	Crampton Lake	2003	197	0.00	22.2
## 42	Crampton Lake	2003	197	0.50	21.9
## 43	Crampton Lake	2003	197	1.00	21.8
## 44	Crampton Lake	2003	197	1.50	21.8
## 45	Crampton Lake	2003	197	2.00	21.7
## 46	Crampton Lake	2003	197	2.50	21.7
## 47	Crampton Lake	2003	197	3.00	21.7
## 48	Crampton Lake	2003	197	3.50	21.6
## 49	Crampton Lake	2003	197	4.00	19.6
## 50	Crampton Lake	2003	197	4.50	17.3
## 51	Crampton Lake	2003	197	5.00	14.5
## 52	Crampton Lake	2003	197	5.50	12.5
## 53	Crampton Lake	2003	197	6.00	10.9
## 54	Crampton Lake	2003	197	6.50	9.8
## 55	Crampton Lake	2003	197	7.00	8.7

## 56	Crampton Lake	2003	197	8.00	7.0
## 57	Crampton Lake	2003	197	9.00	6.0
## 58	Crampton Lake	2003	197	10.00	5.5
## 59	Crampton Lake	2003	197	11.00	5.2
## 60	Crampton Lake	2003	197	12.00	5.1
## 61	Crampton Lake	2003	211	0.00	23.2
## 62	Crampton Lake	2003	211	0.50	23.3
## 63	Crampton Lake	2003	211	1.00	23.2
## 64	Crampton Lake	2003	211	1.50	23.3
## 65	Crampton Lake	2003	211	2.00	23.3
## 66	Crampton Lake	2003	211	2.50	23.3
## 67	Crampton Lake	2003	211	3.00	23.3
## 68	Crampton Lake	2003	211	3.50	23.2
## 69	Crampton Lake	2003	211	4.00	22.4
## 70	Crampton Lake	2003	211	4.50	20.6
## 71	Crampton Lake	2003	211	5.00	17.3
## 72	Crampton Lake	2003	211	5.50	14.6
## 73	Crampton Lake	2003	211	6.00	12.1
## 74	Crampton Lake	2003	211	6.50	11.0
## 75	Crampton Lake	2003	211	7.00	9.6
## 76	Crampton Lake	2003	211	8.00	7.4
## 77	Crampton Lake	2003	211	9.00	6.4
## 78	Crampton Lake	2003	211	10.00	5.7
## 79	Crampton Lake	2003	211	11.00	5.4
## 80	Crampton Lake	2003	211	12.00	5.2
## 81	Crampton Lake	2004	183	0.00	19.8
## 82	Crampton Lake	2004	183	0.50	19.8
## 83	Crampton Lake	2004	183	1.00	19.8
## 84	Crampton Lake	2004	183	1.50	19.8
## 85	Crampton Lake	2004	183	2.00	19.7
## 86	Crampton Lake	2004	183	2.50	19.5
## 87	Crampton Lake	2004	183	3.00	18.9
## 88	Crampton Lake	2004	183	3.50	18.4
## 89	Crampton Lake	2004	183	4.00	18.0
## 90	Crampton Lake	2004	183	4.50	17.5
## 91	Crampton Lake	2004	183	5.00	15.6
## 92	Crampton Lake	2004	183	5.50	13.4
## 93	Crampton Lake	2004	183	6.00	11.2
## 94	Crampton Lake	2004	183	6.50	10.2
## 95	Crampton Lake	2004	183	7.00	9.0
## 96	Crampton Lake	2004	183	8.00	7.6
## 97	Crampton Lake	2004	183	9.00	6.6
## 98	Crampton Lake	2004	183	10.00	6.0
## 99	Crampton Lake	2004	183	11.00	5.6
## 100	Crampton Lake	2004	183	12.00	5.5
## 101	Crampton Lake	2004	183	13.00	5.5
## 102	Crampton Lake	2004	183	14.00	5.5
## 103	Crampton Lake	2004	191	0.00	18.1
## 104	Crampton Lake	2004	191	0.50	18.1
## 105	Crampton Lake	2004	191	1.00	18.0
## 106	Crampton Lake	2004	191	1.50	18.0
## 107	Crampton Lake	2004	191	2.00	17.9
## 108	Crampton Lake	2004	191	2.50	17.9
## 109	Crampton Lake	2004	191	3.00	17.9

## 110	Crampton Lake	2004	191	3.50	17.8
## 111	Crampton Lake	2004	191	4.00	17.6
## 112	Crampton Lake	2004	191	4.50	17.4
## 113	Crampton Lake	2004	191	5.00	16.6
## 114	Crampton Lake	2004	191	5.50	13.8
## 115	Crampton Lake	2004	191	6.00	12.1
## 116	Crampton Lake	2004	191	6.50	10.9
## 117	Crampton Lake	2004	191	7.00	9.4
## 118	Crampton Lake	2004	191	8.00	7.6
## 119	Crampton Lake	2004	191	9.00	6.6
## 120	Crampton Lake	2004	191	10.00	6.0
## 121	Crampton Lake	2004	191	11.00	5.7
## 122	Crampton Lake	2004	191	12.00	5.6
## 123	Crampton Lake	2004	191	13.00	5.5
## 124	Crampton Lake	2004	191	14.00	5.5
## 125	Crampton Lake	2004	197	0.00	21.8
## 126	Crampton Lake	2004	197	0.50	21.8
## 127	Crampton Lake	2004	197	1.00	21.6
## 128	Crampton Lake	2004	197	1.50	21.6
## 129	Crampton Lake	2004	197	2.00	21.5
## 130	Crampton Lake	2004	197	2.50	20.9
## 131	Crampton Lake	2004	197	3.00	19.5
## 132	Crampton Lake	2004	197	3.50	19.1
## 133	Crampton Lake	2004	197	4.00	18.6
## 134	Crampton Lake	2004	197	4.50	17.9
## 135	Crampton Lake	2004	197	5.00	16.7
## 136	Crampton Lake	2004	197	5.50	14.8
## 137	Crampton Lake	2004	197	6.00	12.8
## 138	Crampton Lake	2004	197	6.50	11.1
## 139	Crampton Lake	2004	197	7.00	9.7
## 140	Crampton Lake	2004	197	8.00	7.7
## 141	Crampton Lake	2004	197	9.00	6.6
## 142	Crampton Lake	2004	197	10.00	6.1
## 143	Crampton Lake	2004	197	11.00	5.7
## 144	Crampton Lake	2004	197	12.00	5.6
## 145	Crampton Lake	2004	197	13.00	5.6
## 146	Crampton Lake	2004	197	14.00	5.5
## 147	Crampton Lake	2004	204	0.00	23.8
## 148	Crampton Lake	2004	204	0.50	23.8
## 149	Crampton Lake	2004	204	1.00	23.8
## 150	Crampton Lake	2004	204	1.50	23.8
## 151	Crampton Lake	2004	204	2.00	23.8
## 152	Crampton Lake	2004	204	2.50	23.2
## 153	Crampton Lake	2004	204	3.00	22.0
## 154	Crampton Lake	2004	204	3.50	20.5
## 155	Crampton Lake	2004	204	4.00	19.9
## 156	Crampton Lake	2004	204	4.50	18.5
## 157	Crampton Lake	2004	204	5.00	17.3
## 158	Crampton Lake	2004	204	5.50	15.5
## 159	Crampton Lake	2004	204	6.00	13.5
## 160	Crampton Lake	2004	204	6.50	11.3
## 161	Crampton Lake	2004	204	7.00	10.1
## 162	Crampton Lake	2004	204	8.00	7.8
## 163	Crampton Lake	2004	204	9.00	6.8

## 164	Crampton Lake	2004	204	10.00	6.2
## 165	Crampton Lake	2004	204	11.00	5.8
## 166	Crampton Lake	2004	204	12.00	5.7
## 167	Crampton Lake	2004	204	13.00	5.6
## 168	Crampton Lake	2004	204	14.00	5.6
## 169	Crampton Lake	2004	211	0.00	22.7
## 170	Crampton Lake	2004	211	0.50	22.7
## 171	Crampton Lake	2004	211	1.00	22.7
## 172	Crampton Lake	2004	211	1.50	22.7
## 173	Crampton Lake	2004	211	2.00	22.7
## 174	Crampton Lake	2004	211	2.50	22.7
## 175	Crampton Lake	2004	211	3.00	22.7
## 176	Crampton Lake	2004	211	3.50	22.6
## 177	Crampton Lake	2004	211	4.00	20.8
## 178	Crampton Lake	2004	211	4.50	18.1
## 179	Crampton Lake	2004	211	5.00	17.5
## 180	Crampton Lake	2004	211	5.50	15.8
## 181	Crampton Lake	2004	211	6.00	13.6
## 182	Crampton Lake	2004	211	6.50	11.5
## 183	Crampton Lake	2004	211	7.00	10.3
## 184	Crampton Lake	2004	211	8.00	7.8
## 185	Crampton Lake	2004	211	9.00	6.8
## 186	Crampton Lake	2004	211	10.00	6.2
## 187	Crampton Lake	2004	211	11.00	5.8
## 188	Crampton Lake	2004	211	12.00	5.7
## 189	Crampton Lake	2004	211	13.00	5.6
## 190	Crampton Lake	2004	211	14.00	5.6
## 191	Crampton Lake	2005	186	0.00	22.3
## 192	Crampton Lake	2005	186	0.50	22.3
## 193	Crampton Lake	2005	186	1.00	22.4
## 194	Crampton Lake	2005	186	1.50	22.4
## 195	Crampton Lake	2005	186	2.00	22.4
## 196	Crampton Lake	2005	186	2.50	22.4
## 197	Crampton Lake	2005	186	3.00	22.4
## 198	Crampton Lake	2005	186	3.50	22.3
## 199	Crampton Lake	2005	186	4.00	22.3
## 200	Crampton Lake	2005	186	4.50	21.5
## 201	Crampton Lake	2005	186	5.00	17.9
## 202	Crampton Lake	2005	186	5.50	15.3
## 203	Crampton Lake	2005	186	6.00	13.5
## 204	Crampton Lake	2005	186	6.50	12.3
## 205	Crampton Lake	2005	186	7.00	10.8
## 206	Crampton Lake	2005	186	8.00	8.8
## 207	Crampton Lake	2005	186	9.00	7.4
## 208	Crampton Lake	2005	186	10.00	6.0
## 209	Crampton Lake	2005	186	11.00	5.8
## 210	Crampton Lake	2005	186	12.00	5.7
## 211	Crampton Lake	2005	192	0.00	25.7
## 212	Crampton Lake	2005	192	0.50	25.7
## 213	Crampton Lake	2005	192	1.00	25.4
## 214	Crampton Lake	2005	192	1.50	25.4
## 215	Crampton Lake	2005	192	2.00	25.3
## 216	Crampton Lake	2005	192	2.50	24.2
## 217	Crampton Lake	2005	192	3.00	23.5

##	218	Crampton Lake	2005	192	3.50	22.9
##	219	Crampton Lake	2005	192	4.00	22.5
##	220	Crampton Lake	2005	192	4.50	21.4
##	221	Crampton Lake	2005	192	5.00	19.5
##	222	Crampton Lake	2005	192	5.50	16.5
##	223	Crampton Lake	2005	192	6.00	14.6
##	224	Crampton Lake	2005	192	6.50	12.9
##	225	Crampton Lake	2005	192	7.00	11.4
##	226	Crampton Lake	2005	192	8.00	9.0
##	227	Crampton Lake	2005	192	9.00	7.5
##	228	Crampton Lake	2005	192	10.00	6.5
##	229	Crampton Lake	2005	192	11.00	6.0
##	230	Crampton Lake	2005	192	12.00	5.7
##	231	Crampton Lake	2005	192	13.00	5.6
##	232	Crampton Lake	2005	199	0.00	27.5
##	233	Crampton Lake	2005	199	0.50	27.5
##	234	Crampton Lake	2005	199	1.00	27.5
##	235	Crampton Lake	2005	199	1.50	27.4
##	236	Crampton Lake	2005	199	2.00	27.4
##	237	Crampton Lake	2005	199	2.50	27.4
##	238	Crampton Lake	2005	199	3.00	26.0
##	239	Crampton Lake	2005	199	3.50	24.8
##	240	Crampton Lake	2005	199	4.00	23.7
##	241	Crampton Lake	2005	199	4.50	22.5
##	242	Crampton Lake	2005	199	5.00	19.8
##	243	Crampton Lake	2005	199	5.50	17.1
##	244	Crampton Lake	2005	199	6.00	15.3
##	245	Crampton Lake	2005	199	6.50	13.5
##	246	Crampton Lake	2005	199	7.00	12.0
##	247	Crampton Lake	2005	199	8.00	9.0
##	248	Crampton Lake	2005	199	9.00	7.7
##	249	Crampton Lake	2005	199	10.00	6.6
##	250	Crampton Lake	2005	199	11.00	6.1
##	251	Crampton Lake	2005	199	12.00	5.7
##	252	Crampton Lake	2005	199	13.00	5.7
##	253	Crampton Lake	2005	206	0.00	25.3
##	254	Crampton Lake	2005	206	0.50	25.1
##	255	Crampton Lake	2005	206	1.00	25.0
##	256	Crampton Lake	2005	206	1.50	24.9
##	257	Crampton Lake	2005	206	2.00	24.9
##	258	Crampton Lake	2005	206	2.50	24.9
##	259	Crampton Lake	2005	206	3.00	24.8
##	260	Crampton Lake	2005	206	3.50	24.8
##	261	Crampton Lake	2005	206	4.00	24.7
##	262	Crampton Lake	2005	206	4.50	24.5
##	263	Crampton Lake	2005	206	5.00	21.3
##	264	Crampton Lake	2005	206	5.50	18.8
##	265	Crampton Lake	2005	206	6.00	16.7
##	266	Crampton Lake	2005	206	6.50	14.4
##	267	Crampton Lake	2005	206	7.00	12.3
##	268	Crampton Lake	2005	206	8.00	9.9
##	269	Crampton Lake	2005	206	9.00	7.9
##	270	Crampton Lake	2005	206	10.00	6.8
##	271	Crampton Lake	2005	206	11.00	6.1

##	272	Crampton Lake	2005	206	12.00	5.8
##	273	Crampton Lake	2005	206	13.00	5.7
##	274	Crampton Lake	2006	186	0.00	22.7
##	275	Crampton Lake	2006	186	0.50	22.8
##	276	Crampton Lake	2006	186	1.00	22.8
##	277	Crampton Lake	2006	186	1.50	22.8
##	278	Crampton Lake	2006	186	2.00	22.8
##	279	Crampton Lake	2006	186	2.50	22.5
##	280	Crampton Lake	2006	186	3.00	22.6
##	281	Crampton Lake	2006	186	3.50	22.6
##	282	Crampton Lake	2006	186	4.00	20.3
##	283	Crampton Lake	2006	186	4.50	17.4
##	284	Crampton Lake	2006	186	5.00	15.2
##	285	Crampton Lake	2006	186	5.50	13.7
##	286	Crampton Lake	2006	186	6.00	12.1
##	287	Crampton Lake	2006	186	6.50	10.8
##	288	Crampton Lake	2006	186	7.00	9.6
##	289	Crampton Lake	2006	186	7.50	8.3
##	290	Crampton Lake	2006	186	8.00	7.4
##	291	Crampton Lake	2006	186	8.50	6.3
##	292	Crampton Lake	2006	186	9.00	6.0
##	293	Crampton Lake	2006	186	10.00	5.5
##	294	Crampton Lake	2006	186	11.00	5.2
##	295	Crampton Lake	2006	186	12.00	5.1
##	296	Crampton Lake	2006	200	0.00	25.4
##	297	Crampton Lake	2006	200	0.50	25.4
##	298	Crampton Lake	2006	200	1.00	25.4
##	299	Crampton Lake	2006	200	1.50	25.4
##	300	Crampton Lake	2006	200	2.00	25.4
##	301	Crampton Lake	2006	200	2.50	25.4
##	302	Crampton Lake	2006	200	3.00	25.4
##	303	Crampton Lake	2006	200	3.50	24.1
##	304	Crampton Lake	2006	200	4.00	22.5
##	305	Crampton Lake	2006	200	4.50	19.8
##	306	Crampton Lake	2006	200	5.00	17.5
##	307	Crampton Lake	2006	200	5.50	15.0
##	308	Crampton Lake	2006	200	6.00	13.3
##	309	Crampton Lake	2006	200	6.50	11.4
##	310	Crampton Lake	2006	200	7.00	9.6
##	311	Crampton Lake	2006	200	7.50	8.7
##	312	Crampton Lake	2006	200	8.00	7.5
##	313	Crampton Lake	2006	200	8.50	6.7
##	314	Crampton Lake	2006	200	9.00	6.4
##	315	Crampton Lake	2006	200	10.00	5.7
##	316	Crampton Lake	2006	200	11.00	5.4
##	317	Crampton Lake	2006	200	12.00	5.1
##	318	Crampton Lake	2006	200	13.00	5.1
##	319	Ward Lake	2010	188	0.00	25.5
##	320	Ward Lake	2010	188	0.50	25.3
##	321	Ward Lake	2010	188	0.75	24.9
##	322	Ward Lake	2010	188	1.00	24.8
##	323	Ward Lake	2010	188	1.50	23.7
##	324	Ward Lake	2010	188	2.00	21.8
##	325	Ward Lake	2010	188	2.50	18.6

## 326	Ward Lake	2010	188	3.00	16.6
## 327	Ward Lake	2010	188	3.50	14.6
## 328	Ward Lake	2010	188	4.00	12.5
## 329	Ward Lake	2010	188	4.50	10.4
## 330	Ward Lake	2010	188	5.00	9.0
## 331	Ward Lake	2010	188	5.50	7.7
## 332	Ward Lake	2010	188	6.00	7.1
## 333	Ward Lake	2010	188	6.50	6.9
## 334	Ward Lake	2010	188	6.75	6.7
## 335	Ward Lake	2010	195	0.00	24.1
## 336	Ward Lake	2010	195	0.50	24.1
## 337	Ward Lake	2010	195	1.00	24.0
## 338	Ward Lake	2010	195	1.50	23.9
## 339	Ward Lake	2010	195	2.00	23.2
## 340	Ward Lake	2010	195	2.50	20.3
## 341	Ward Lake	2010	195	3.00	17.4
## 342	Ward Lake	2010	195	3.50	14.7
## 343	Ward Lake	2010	195	4.00	12.5
## 344	Ward Lake	2010	195	4.50	10.6
## 345	Ward Lake	2010	195	5.00	8.8
## 346	Ward Lake	2010	195	5.50	7.7
## 347	Ward Lake	2010	195	6.00	7.2
## 348	Ward Lake	2010	195	6.50	7.0
## 349	Ward Lake	2010	202	0.00	23.9
## 350	Ward Lake	2010	202	0.50	23.8
## 351	Ward Lake	2010	202	1.00	23.7
## 352	Ward Lake	2010	202	1.50	23.3
## 353	Ward Lake	2010	202	2.00	23.1
## 354	Ward Lake	2010	202	2.50	20.7
## 355	Ward Lake	2010	202	3.00	17.0
## 356	Ward Lake	2010	202	3.50	15.1
## 357	Ward Lake	2010	202	4.00	12.6
## 358	Ward Lake	2010	202	4.50	10.7
## 359	Ward Lake	2010	202	5.00	9.1
## 360	Ward Lake	2010	202	5.50	7.9
## 361	Ward Lake	2010	202	6.00	7.3
## 362	Ward Lake	2010	202	6.50	7.1
## 363	Ward Lake	2010	202	7.00	7.1
## 364	Ward Lake	2010	209	0.00	23.6
## 365	Ward Lake	2010	209	0.50	23.6
## 366	Ward Lake	2010	209	1.00	23.5
## 367	Ward Lake	2010	209	1.50	23.4
## 368	Ward Lake	2010	209	2.00	23.1
## 369	Ward Lake	2010	209	2.50	21.0
## 370	Ward Lake	2010	209	3.00	18.2
## 371	Ward Lake	2010	209	3.50	15.7
## 372	Ward Lake	2010	209	4.00	13.4
## 373	Ward Lake	2010	209	4.50	11.1
## 374	Ward Lake	2010	209	5.00	9.2
## 375	Ward Lake	2010	209	5.50	8.0
## 376	Ward Lake	2010	209	6.00	7.4
## 377	Ward Lake	2010	209	6.50	7.2
## 378	Ward Lake	2010	209	6.75	7.0
## 379	Ward Lake	2012	187	0.00	27.6

## 380	Ward Lake	2012	187	0.50	27.0
## 381	Ward Lake	2012	187	1.00	24.6
## 382	Ward Lake	2012	187	1.50	20.4
## 383	Ward Lake	2012	187	2.00	15.8
## 384	Ward Lake	2012	187	2.50	12.3
## 385	Ward Lake	2012	187	3.00	10.2
## 386	Ward Lake	2012	187	3.50	7.9
## 387	Ward Lake	2012	187	4.00	6.9
## 388	Ward Lake	2012	187	4.50	6.4
## 389	Ward Lake	2012	187	5.00	6.1
## 390	Ward Lake	2012	187	5.50	5.8
## 391	Ward Lake	2012	187	6.00	5.7
## 392	Ward Lake	2012	194	0.00	25.9
## 393	Ward Lake	2012	194	0.50	25.1
## 394	Ward Lake	2012	194	1.00	24.5
## 395	Ward Lake	2012	194	1.50	20.8
## 396	Ward Lake	2012	194	2.00	16.5
## 397	Ward Lake	2012	194	2.50	12.6
## 398	Ward Lake	2012	194	3.00	10.3
## 399	Ward Lake	2012	194	3.50	8.2
## 400	Ward Lake	2012	194	4.00	7.2
## 401	Ward Lake	2012	194	4.50	6.6
## 402	Ward Lake	2012	194	5.00	6.3
## 403	Ward Lake	2012	194	5.50	6.0
## 404	Ward Lake	2012	194	6.00	5.9
## 405	Ward Lake	2012	201	0.00	23.6
## 406	Ward Lake	2012	201	0.50	23.3
## 407	Ward Lake	2012	201	1.00	23.2
## 408	Ward Lake	2012	201	1.50	21.5
## 409	Ward Lake	2012	201	2.00	16.9
## 410	Ward Lake	2012	201	2.50	13.2
## 411	Ward Lake	2012	201	3.00	10.3
## 412	Ward Lake	2012	201	3.50	8.6
## 413	Ward Lake	2012	201	4.00	7.7
## 414	Ward Lake	2012	201	4.50	6.8
## 415	Ward Lake	2012	201	5.00	6.4
## 416	Ward Lake	2012	201	5.50	6.2
## 417	Ward Lake	2012	201	6.00	6.0
## 418	Ward Lake	2012	201	6.50	5.9
## 419	Ward Lake	2012	201	7.00	5.8
## 420	Ward Lake	2012	208	0.00	24.3
## 421	Ward Lake	2012	208	0.50	24.2
## 422	Ward Lake	2012	208	1.00	24.2
## 423	Ward Lake	2012	208	1.50	22.3
## 424	Ward Lake	2012	208	2.00	17.5
## 425	Ward Lake	2012	208	2.50	13.7
## 426	Ward Lake	2012	208	3.00	10.5
## 427	Ward Lake	2012	208	3.50	8.5
## 428	Ward Lake	2012	208	4.00	7.6
## 429	Ward Lake	2012	208	4.50	6.8
## 430	Ward Lake	2012	208	5.00	6.4
## 431	Ward Lake	2012	208	5.50	6.1
## 432	Ward Lake	2012	208	6.00	6.0
## 433	Ward Lake	2012	208	6.50	5.9

```
## 434      Ward Lake  2012    208  7.00          5.8
```

```
tail(ChemPhys_Crampton_Ward)
```

```
##      lakename year4 daynum depth temperature_C
## 429 Ward Lake  2012    208   4.5          6.8
## 430 Ward Lake  2012    208   5.0          6.4
## 431 Ward Lake  2012    208   5.5          6.1
## 432 Ward Lake  2012    208   6.0          6.0
## 433 Ward Lake  2012    208   6.5          5.9
## 434 Ward Lake  2012    208   7.0          5.8
```

```
Crampton_Ward_ttest <- t.test(ChemPhys_Crampton_Ward$temperature_C
                             ~ ChemPhys_Crampton_Ward$lakename)
Crampton_Ward_ttest
```

```
##
## Welch Two Sample t-test
##
## data: ChemPhys_Crampton_Ward$temperature_C by ChemPhys_Crampton_Ward$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129  2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##           15.35189              14.45862
```

Answer: This test says that the p-value for this relationship is 0.2649. The mean temperature for Crampton Lake is 15.35 degrees C and the mean temperature for Ward Lake is 14.45 degrees C (unequal). This result does not match my answer for part 16. The post-hoc test found that the p-value between Crampton Lake and Ward Lake was 0.9714459. However, both values are statistically insignificant, so in practice, this doesn't change our acceptance of the null hypothesis in terms of the mean temperature of these two lakes. In other words, the mean July temperature of Crampton and Ward Lake is the same.